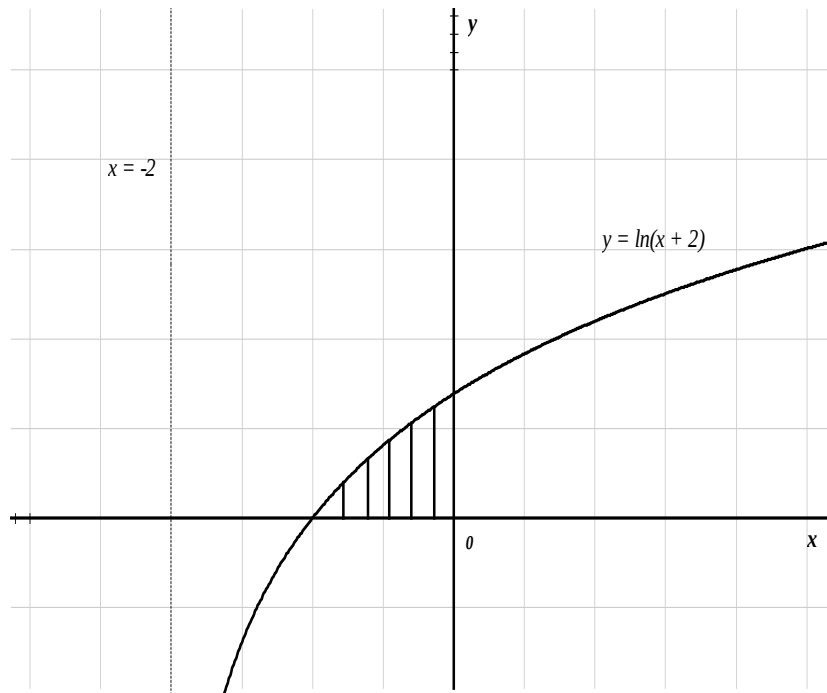


Question One**Marks**

- a) Find $\int \sin(4x+6)dx$. **1**
- b) Differentiate 5^x . **1**
- c) For what values of x will $1 - \tan^2 x + \tan^4 x - \tan^6 x + \tan^8 x - \dots$ have a limiting sum for $0 \leq x \leq 2\pi$. **4**
- d) The sketch of the curve of $y = \ln(x+2)$ is shown below. If the shaded area is rotated about the y -axis, find the volume of revolution of the solid generated. **4**

**Question Two (Start a New Page)**

- a) i) Express $y = \sqrt{3} \cos x - \sin x$ in the form of $R \cos(x - \alpha)$, where $R > 0$ and $0 \leq \alpha \leq 2\pi$. **3**
- ii) Sketch the graph of $y = \sqrt{3} \cos x - \sin x$, for $0 \leq x \leq 2\pi$. **3**
- b) i) Differentiate $\ln(\sin x) - x \cot x$. **2**
- ii) Hence find $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} x(\operatorname{cosec}^2 x) dx$. **2**

Question Three (Start a New Page)**Marks**

- a) Prove that $2^{10n+3} + 3$ is divisible by 11 for all non-negative integers n by Mathematical Induction. **5**
- b) A spherical map of the earth is being inflated at a constant rate of $25\text{cm}^3\text{s}^{-1}$. Find the rate at which the length of the equator is changing when the radius is 10cm. **5**

Question Four (Start a New Page)

- a) Differentiate $\ln\left(\frac{\sqrt{x}}{x+4}\right)$. **2**
- b) Consider the function $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$,
- i) Prove that y has no stationary points. **2**
- ii) Prove that the lines $y = \pm 1$ are asymptotes. **2**
- iii) If k is a positive constant, find the area in the first quadrant enclosed by the above curve and the three lines $y=1$, $x=0$ and $x=k$. **3**
- iv) Prove that for all values of k , the area is always less than $\ln 2$. **1**

Question Five (Start a New Page)

- a) Find $\int \frac{x-1}{x+5} dx$. **2**
- b) Evaluate $\int_0^{\ln\sqrt{3}} \frac{e^x}{1+e^{2x}} dx$. **3**
- c) Amy borrows \$130 000 to start a sign writing business. Interest is charged on the balance owing at the rate of 9% per annum, compounded monthly. Amy agrees to repay the loan, including interest, by making equal monthly instalment of \$ P .
- i) How much does Amy owe at the end of the first month just before she makes an instalment payment? **1**

Question Five cont'd**Marks**

- c) ii) Show that if the loan is repaid after n months, then

1

$$P = \frac{130000(1.0075)^n}{1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1}}.$$

- iii) Calculate how many months, to the nearest month, it will take for the loan to be repaid if Amy makes instalments of \$1800 per month.

3**Question Six (Start a New Page)**

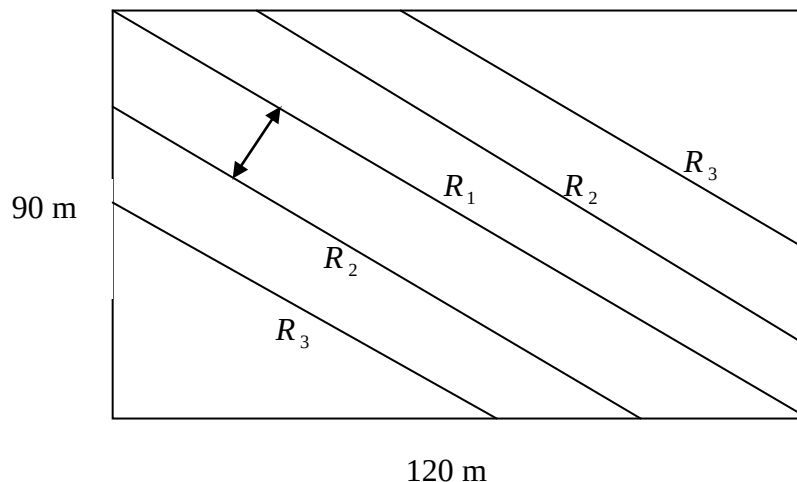
- a) Using the substitution $x = 2 \sin \theta$, show by integration that

4

$$\int \sqrt{4-x^2} \, dx = \frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2} + C, \text{ where } C \text{ is a constant.}$$

- b) A rectangular paddock in a vineyard measures 90m by 120m. In order to make best use of the sun, the grape vines are planted in diagonal rows as shown, with a 3 metre gap between adjacent rows.

Diagram not to scale



- i) Find the length of R_1 , the diagonal of the field.
- ii) Show that length of the R_2 is 143.75 m.
- iii) Given that the rows $R_1 + R_2 + R_3 + R_4 + \dots$ form an arithmetic series, find the total number of rows of vines in the paddock.

1**2****3****END OF PAPER**

Question One

a) $\frac{-\cos(4x+b)}{4} + c$

b) $5^x(\ln 5)$

c) $r = -\tan^2 x$

$|r| < 1$ when $\tan^2 x < 1$

$-1 < \tan x < 1$

$\therefore 0 < x < \frac{\pi}{4}, \quad \frac{3\pi}{4} < x < \frac{5\pi}{4}, \quad \frac{7\pi}{4} < x < 2\pi$
 $x \neq \frac{\pi}{2}, \frac{3\pi}{2}$

d) vol of revolution (V)

$$= \int_0^{\ln 2} \pi x^2 dy$$

$$= \pi \int_0^{\ln 2} (e^y - 2)^2 dy$$

$$= \pi \int_0^{\ln 2} (e^{2y} - 4e^y + 4) dy$$

$$= \pi \left[\frac{e^{2y}}{2} - 4e^y + 4y \right]_0^{\ln 2}$$

$$= \pi \left[\frac{4}{2} - 4 \cdot 2 + 4 \ln^2 2 \right] - \pi \left[\frac{1}{2} - 4 \right]$$

$$= \pi \left[4 \ln^2 2 - 2 \frac{1}{2} \right] \text{ unit}^3$$

Question Two

a) $R \cos(x-\alpha) = R \left[\cos x \cos \alpha + \sin x \sin \alpha \right]$

$$= \sqrt{3} \cos x - \sin x$$

$$\begin{aligned} R > 0, \quad \sqrt{3} &= R \cos \alpha \\ -1 &= R \sin \alpha \end{aligned} \quad \therefore \alpha \text{ in 4th Quad}$$

$$\tan \alpha = -\frac{1}{\sqrt{3}} \quad / \quad \alpha = \frac{11\pi}{6}$$

$$(R \cos \alpha) + (R \sin \alpha) = (\sqrt{3}) + (-1)$$

$$R^2 = 4 \quad \therefore R = 2 \quad (R > 0)$$

$$\therefore \sqrt{3} \cos x - \sin x = 2 \cos \left(x - \frac{11\pi}{6} \right)$$

$$y = 2 \cos \left(x - \frac{11\pi}{6} \right) \quad \text{Period} : 2\pi$$

At $x=0$ $y = 2 \cos \left(-\frac{11\pi}{6} \right) = \sqrt{3}$

$y=0$ when $\cos \left(x - \frac{11\pi}{6} \right) = 0$

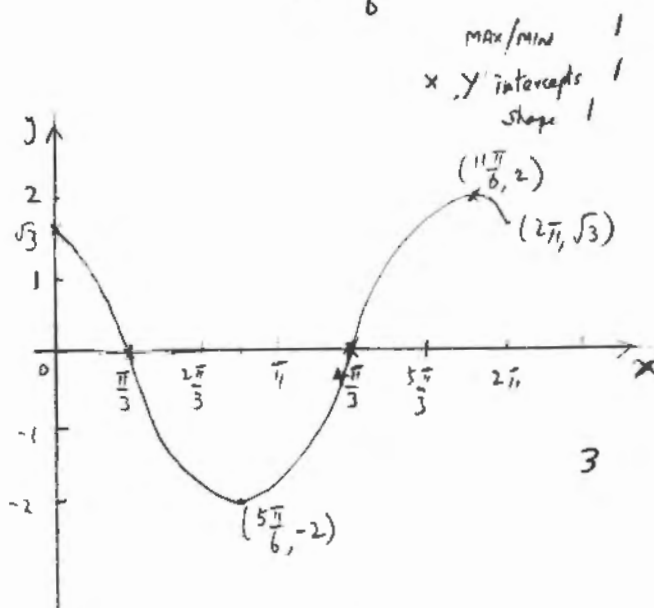
$x - \frac{11\pi}{6} = -\frac{\pi}{2} \quad \therefore x = \frac{4\pi}{3}$

$x - \frac{11\pi}{6} = -\frac{3\pi}{2} \quad x = \frac{\pi}{3}$

MAX at $y=2$ $2 = 2 \cos \left(x - \frac{11\pi}{6} \right)$

$0 = x - \frac{11\pi}{6}$

$x = \frac{11\pi}{6}$



Let $y = \ln(\sin x) - x \cot x$

b) $y' = \frac{\cos x}{\sin x} + x \csc^2 x - \cot x$

$$= \cot x + x \csc^2 x - \cot x$$

$$y' = x \csc^2 x$$

ii) $\int_{\pi/6}^{\pi/2} x \csc^2 x dx = \left[\ln |\sin x| - x \cot x \right]_{\pi/6}^{\pi/2}$

$$= \left(\ln \left| \sin \frac{\pi}{2} \right| - \frac{\pi}{2} \cot \frac{\pi}{2} \right) - \left(\ln \left| \sin \frac{\pi}{6} \right| - \frac{\pi}{6} \cot \frac{\pi}{6} \right)$$

$$= \left(\ln 1 - \frac{\pi}{2} \cdot 0 \right) - \left(\ln \frac{1}{2} - \frac{\pi}{6} \sqrt{3} \right)$$

$$= -\ln \frac{1}{2} + \frac{\pi}{6} \sqrt{3} = \frac{\pi}{6} \sqrt{3} + \ln 2$$

Question 3

a) When $n=0$ $2^{10 \cdot 0 + 3} + 3 = 2^3 + 3 = 11$ divisible by 11

Assume $2^{10k+3} + 3$ is divisible by 11

i.e. $2^{10k+3} + 3 = 11Q$ $Q \in \mathbb{Z}^+$

$\therefore 2^{10k+3} = 11Q - 3$

Required to prove $2^{10(k+1)+3} + 3$ is divisible by 11

$2^{10(k+1)+3} + 3 = (2^{10k+3}) \cdot 2^{10} + 3$

$= (11Q - 3) \cdot 2^{10} + 3$ from assumption

$= 11Q \cdot 2^{10} - 3 \cdot 2^{10} + 3$

$= 11Q \cdot 2^{10} - 3072 + 3$

$= 11Q \cdot 2^{10} - 3069$

$= 11[1024Q - 279]$

$2^{10}Q - 279$ is a positive integer

Since $Q \in \mathbb{Z}^+$

$\therefore 2^{10(k+1)+3} + 3$ is divisible by 11

Hence by the Principle of Mathematical Induction $2^{10n+3} + 3$ is divisible by all non-negative integers n .

b) $V = \frac{4}{3}\pi r^3$ (vol. of sphere)

length of equation $l = 2\pi r \therefore r = \frac{l}{2\pi}$

$V = \frac{4}{3}\pi \left(\frac{l}{2\pi}\right)^3 = \frac{l^3}{6\pi^2}$

$\frac{dV}{dl} = \frac{3l^2}{6\pi^2} = \frac{l^2}{2\pi^2}$

Given $\frac{dV}{dt} = 25$

$\frac{dR}{dt} = \frac{dl}{dV} \cdot \frac{dV}{dt} = \frac{l^2}{2\pi^2} \times 25 = \frac{50\pi^2}{l^2}$

when $r=10$ $l=20\pi$

$\therefore \frac{dl}{dt} = \frac{50\pi^2}{400\pi^2} = \frac{1}{8} \text{ cm/s}$

Question 4

a) $\ln\left(\frac{\sqrt{x}}{x+4}\right) = \frac{1}{2}\ln x - \ln(x+4)$

$\frac{d}{dx} \left(\ln \frac{\sqrt{x}}{x+4} \right) = \frac{1}{2x} - \frac{1}{x+4}$

$= \frac{x+4-2x}{2x(x+4)}$

$= \frac{4-x}{2x(x+4)}$

b) i) $y' = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$

$y = \frac{e^{2x} + 1 + 1 + e^{-2x} - (e^{2x} - 1 - 1 + e^{-2x})}{(e^x + e^{-x})^2}$

$y' = \frac{4}{(e^x + e^{-x})^2} \neq 0$

$\therefore$ No stationary point

ii) $y = \frac{(e^x - e^{-x}) \div e^x}{(e^x + e^{-x}) \div e^x}$

$y = \frac{1 - e^{-2x}}{1 + e^{-2x}}$

as $x \rightarrow \infty$ $e^{-2x} \rightarrow 0 \therefore y \rightarrow 1$

Similarly $y = \frac{(e^x - e^{-x}) \div e^{-x}}{(e^x + e^{-x}) \div e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$

as $x \rightarrow -\infty$ $e^{2x} \rightarrow 0 \therefore y \rightarrow -1$

Hence the lines $y = \pm 1$ are asymptotes

iii) Shaded Area A

$= k \times 1 - \int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$

$= k - \ln|e^x + e^{-x}| \Big|_0^k$

$= k - \ln(e^k + e^{-k}) + \ln(1+1)$

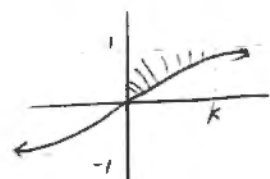
$= k - \ln(e^k + e^{-k}) + \ln 2$

iv) since $e^k > 0$ $e^{-k} > 0$

$e^k < e^k + e^{-k}$

Taking loge on both side $k < \ln(e^k + e^{-k})$

$\therefore A < \ln 2$



Question 5

$$a) \int \frac{x-1}{x+5} dx = \int \frac{x+5-6}{x+5} dx = \int \left(1 - \frac{6}{x+5} \right) dx$$

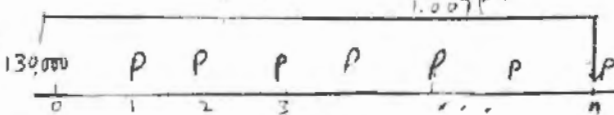
$$= x - 6 \ln(x+5) + C$$

$$b) \int_0^{\ln \sqrt{3}} \frac{e^x}{1+e^{2x}} dx = \int_1^{\sqrt{3}} \frac{u}{1+u^2} du \quad \begin{matrix} u=e^x \\ du=e^x dx \\ \text{when } x=0, u=1 \\ \text{when } x=\ln \sqrt{3}, u=\sqrt{3} \end{matrix}$$

$$= \left[\frac{1}{2} \ln(1+u^2) \right]_1^{\sqrt{3}} = \frac{1}{2} \ln 4 - \frac{1}{2} \ln 2 = \frac{1}{2} \ln 2$$

c) 9% p.a. = 0.75% per month

i) At the end of first month = $130000(1.0075) - P$

ii) 

At the end of n months the loan of \$130,000 will accumulate to $130000(1.0075)^n$ and the sum of all instalments will accumulate to

$$P(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1})$$

For the loan to be paid off

$$130000(1.0075)^n = P(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1})$$

$$\therefore P = \frac{130000(1.0075)^n}{1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1}}$$

iii) $1800 = \frac{130000(1.0075^n)}{1 + 1.0075 + \dots + 1.0075^{n-1}} \quad (\text{from ii})$

$$1800 = \frac{130000(1.0075^n)}{\frac{1 - 1.0075^n}{1 - 1.0075}}$$

$$240000 \times (1.0075^n - 1) = 130000(1.0075^n)$$

$$24(1.0075^n) - 24 = 13(1.0075^n)$$

$$11(1.0075^n) = 24$$

$$n(\log 1.0075) = \log \left(\frac{24}{11} \right)$$

$$n \approx 104 \quad (\text{nearest month})$$

Question 6

a) $x = 2 \sin \theta \quad \frac{dx}{d\theta} = 2 \cos \theta$

$$\int \sqrt{4-x^2} dx = \int \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta$$

$$= \int (2 \cos \theta)(2 \cos \theta) d\theta$$

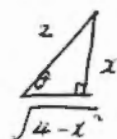
$$= 4 \int \cos^2 \theta d\theta = 4 \int \frac{1 + \cos 2\theta}{2} d\theta$$

$$= 2 \left(\theta + \frac{\sin 2\theta}{2} \right) + C$$

$$= 2 \sin^{-1} \left(\frac{x}{2} \right) + 2 \sin \theta \cos \theta + C$$

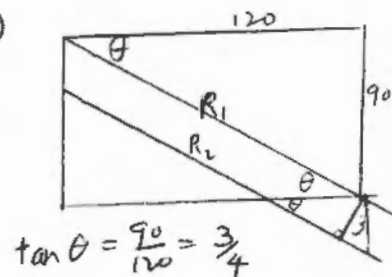
$$= 2 \sin^{-1} \left(\frac{x}{2} \right) + x \sqrt{4-x^2} + C$$

$$= \frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) + C$$

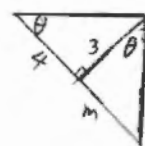


b) $R_1 = \sqrt{120^2 + 90^2} = 150 \text{ m}$

ii)



$$\tan \theta = \frac{90}{120} = \frac{3}{4}$$



$$\tan \theta = \frac{3}{4} = \frac{m}{n}$$

$$4n = 9$$

$$n = 2.25$$

$$\therefore R_2 = 150 - 2.25 \cdot 4 = 143.75 \text{ m}$$

iii) Cut the rectangle into 2 halves (triangles). Let n be the last row in the triangle.

$$R_n = a + (n-1)d = 150 - 6.25(n-1) \geq 0$$

$$150 \geq 6.25(n-1)$$

$$24 \geq n-1$$

$$n \leq 25 \quad \therefore n = 24$$

Total number of rows of vines

$$= 23 + 24 = 47$$